Lab 2

Objective:

The purpose of this lab was to explore the use of operational amplifiers (op-amps) in control systems, particularly focusing on comparator circuits with hysteresis and binary-to-analog conversion circuits.

1. Op-Amps in Control Systems

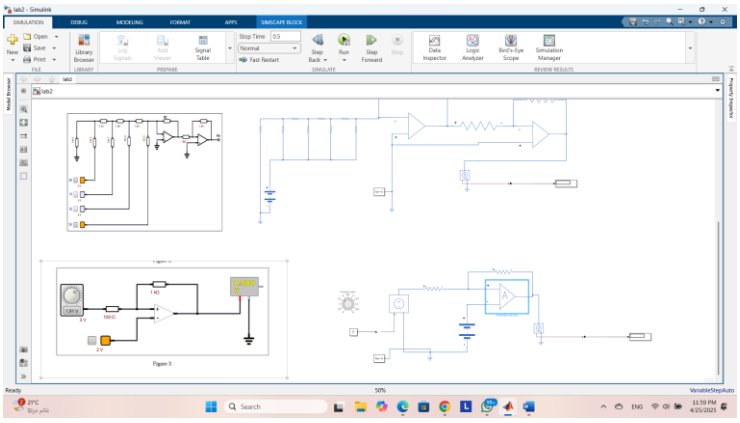
Operational amplifiers (op-amps) are fundamental building blocks in analog electronics. In this lab, op-amps were used in open-loop and feedback configurations. Their high gain and differential input make them suitable for signal conditioning, filtering, and control logic. We mainly used the LM741 op-amp to build and test various control circuits.

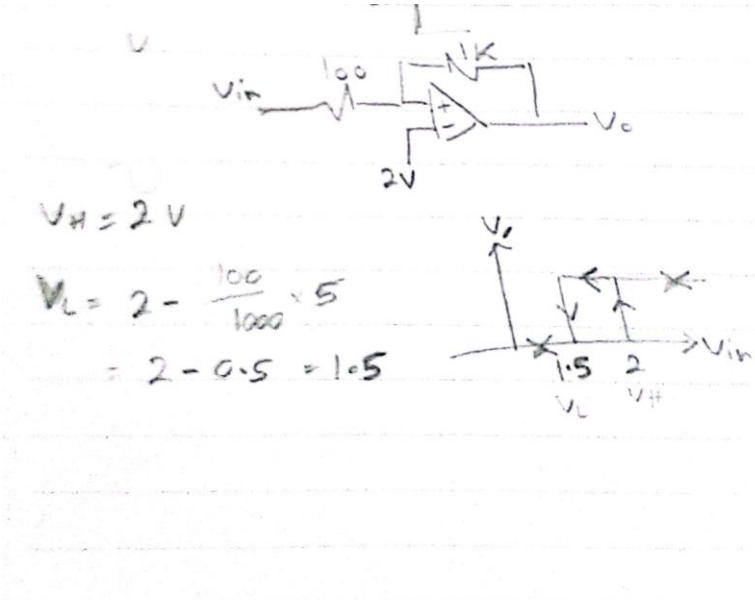
2. Hysteresis Comparator Circuit

The hysteresis comparator circuit, also known as a Schmitt trigger, was designed to compare an input voltage to a reference threshold and provide a clean digital output. The hysteresis effect was introduced using positive feedback through a resistor connected from the output back to the non-inverting input.

This design ensures that small fluctuations in the input signal do not cause rapid switching at the output. The output only switches when the input crosses defined upper and lower threshold voltages, improving noise immunity.

- Key observations:
- When the input voltage rose above the upper threshold, the output snapped high.
- When the input fell below the lower threshold, the output snapped low.
- The width of the hysteresis window was determined by the feedback and input resistor values.



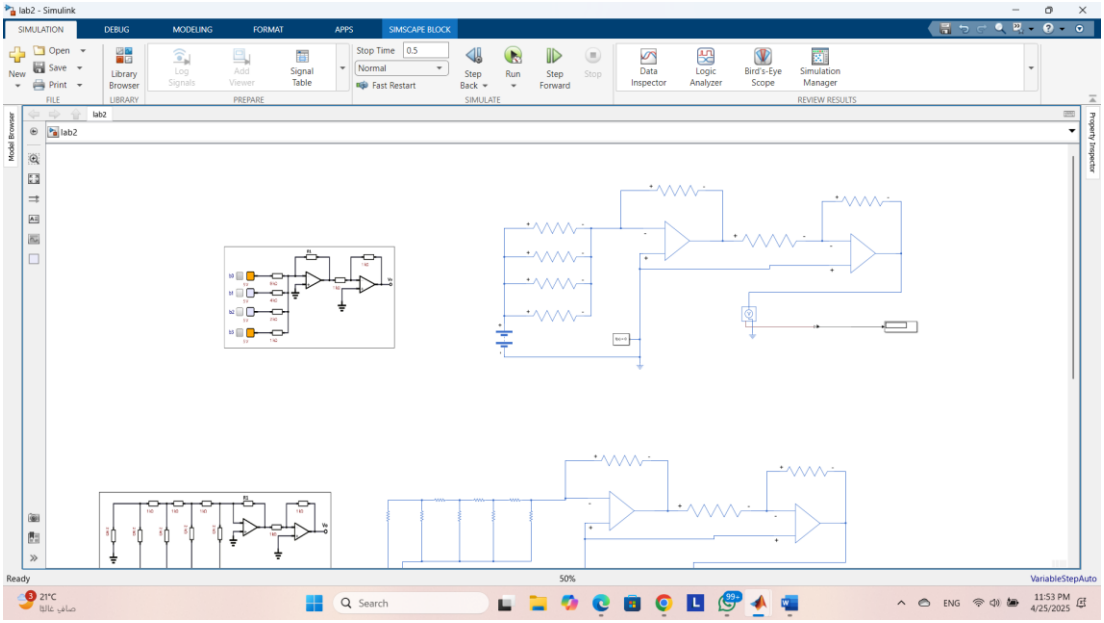


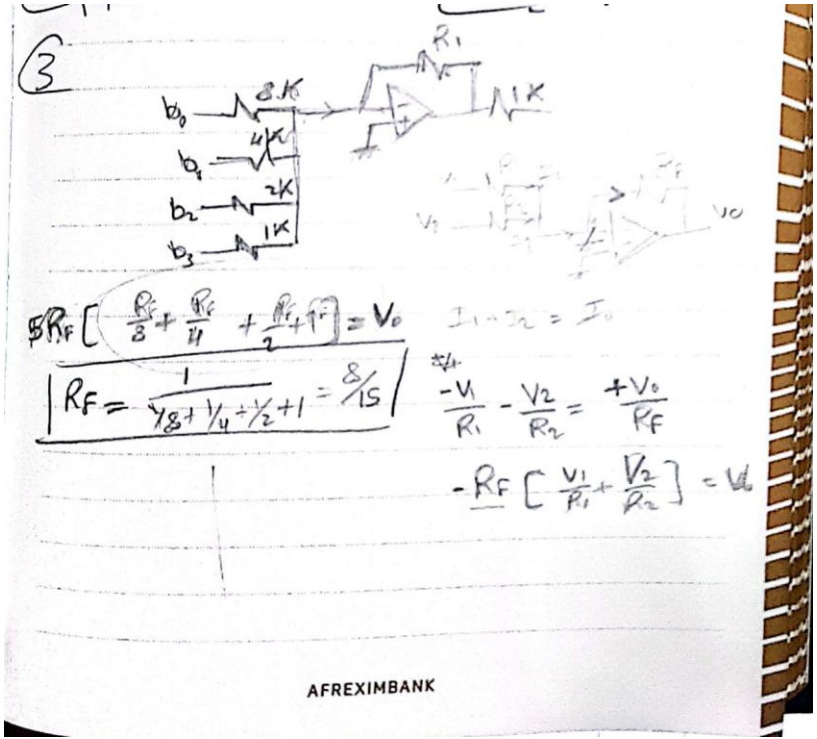
3. Binary-to-Analog Converter (DAC)

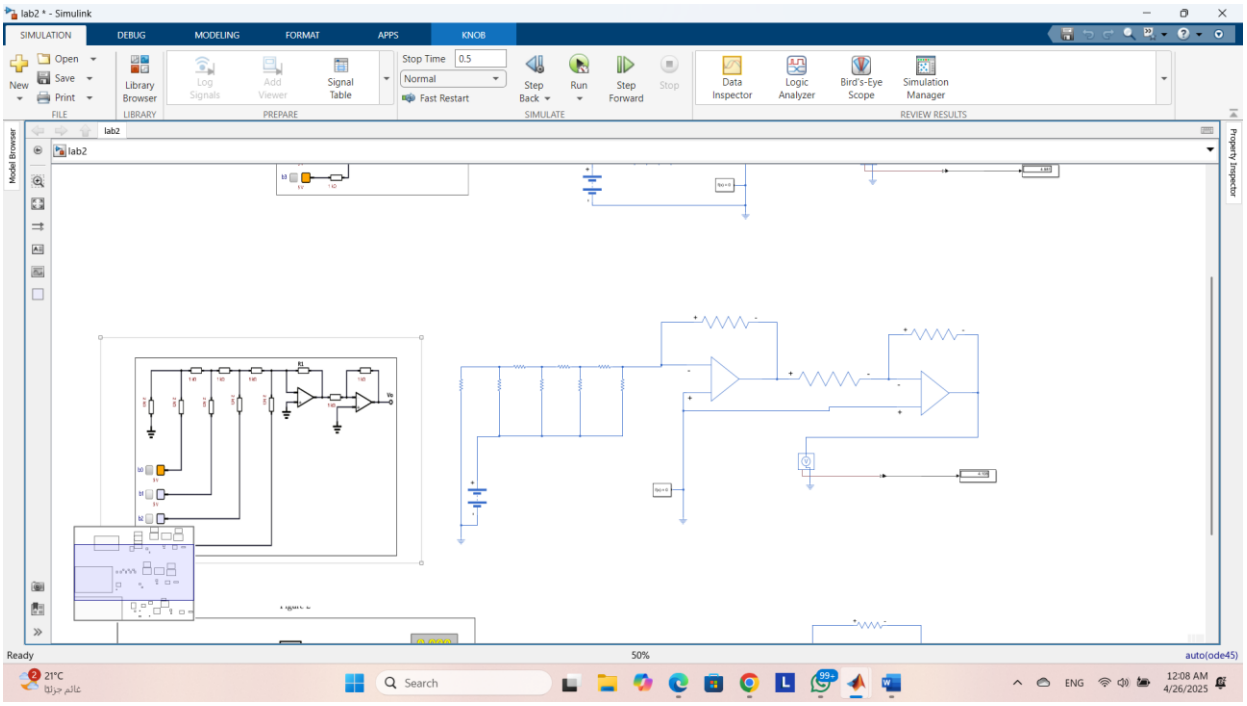
A basic binary-to-analog circuit was built using a weighted resistor network and an op-amp in summing configuration. Digital input bits (e.g., from a DIP switch or function generator) were applied, and the output voltage represented a proportional analog value.

The DAC circuit converted 4-bit binary inputs into corresponding voltage levels. The output was observed to increase in discrete steps as the binary input increased.

- Key observations:
- The output voltage was linear with respect to the binary input.
- Precision of the output depended on accurate resistor values.
- A buffering op-amp ensured that the output was not affected by loading.






lab2 * - Simulink
SIMULATION
DEBUG
MODELING
FORMAT
APPS
KNOB
New
Open
Save
Print
Library Browser
Log Signals
Add Viewer
Signal Table
Stop Time
0.5
Normal
Fast Restart
Step Back
Run
Step Forward
Stop
Data Inspector
Logic Analyzer
Bird's-Eye Scope
Simulation Manager
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